ELEC2104/9704 tutorial 3 (2025)

Before the tutorial, there is a small formula sheet for you.

V_{bi} **or** ϕ_{bi} **or** ϕ_j Built-in potential across the depletion region at zero applied bias (V).

$E_{\max}$ Maximum magnitude of the electric field at the metallurgical junction (V/m or V/µm).

W Total depletion width, $W = x_p + x_n$ (m or µm).

x_p Portion of the depletion width extending into the p-type side (m or µm).

x_n Portion of the depletion width extending into the n-type side (m or µm).

q Elementary charge, $1.602 \times 10^{-19}\,\mathrm{C}$.

N_A Acceptor (p-type) doping concentration (cm^{-3} or m^{-3}).

N_D Donor (n-type) doping concentration (cm^{-3} or m^{-3}).

ε_s Semiconductor permittivity ($\varepsilon_s = \varepsilon_r \varepsilon_0$; for Si, $\varepsilon_r \approx 11.7$).

ε_0 Vacuum permittivity, $8.854 \times 10^{-12}\,\mathrm{F/m}$.

k_B Boltzmann constant, $1.380\,649 \times 10^{-23}\,\mathrm{J/K} = 8.617\,333 \times 10^{-5}\,\mathrm{eV/K}$.

T Absolute temperature (K).

n_i Intrinsic carrier concentration of the semiconductor (cm^{-3} or m^{-3}).

$E_{i,p}$, $E_{i,n}$ Intrinsic Fermi levels in the p- and n-side quasi-neutral regions (eV); at equilibrium

There are several formulas for the built-in voltage; you can pick any of them regarding the provided parameters that are known.

$$(1)\quad V_{bi} = \tfrac{1}{2}\, E_{\max}\, W \tag{1}$$

$$(2)\quad V_{bi} = \tfrac{1}{2}\frac{qN_A}{\varepsilon_s}\, x_p^2 + \tfrac{1}{2}\frac{qN_D}{\varepsilon_s}\, x_n^2 \tag{2}$$

$$(3)\quad V_{bi} = \tfrac{1}{2}\frac{q}{\varepsilon_s}\,\frac{N_A N_D}{N_A + N_D}\, W^2 \tag{3}$$

$$(4)\quad V_{bi} = \frac{1}{q}\,(E_{i,p} - E_{i,n}) = \frac{k_B T}{q}\,\ln\!\left(\frac{N_A N_D}{n_i^2}\right) \tag{4}$$

1 Question 1

We have a p-type semiconductor with a dopant concentration $N_A = 10^{17}\,\mathrm{cm}^{-3}$. Given the intrinsic carrier concentration $n_i = 10^{10}\,\mathrm{cm}^{-3}$, $k_B T = 26\,\mathrm{meV}$, where is the Fermi level with respect to E_i?

Solution:
For a p-type silicon, the acceptor density N_A is much higher than the donor density N_D. Since N_D is

not given, we assume it is negligible, that is close to $0\,\mathrm{cm}^{-3}$. Then we can calculate the hole density p by:

$$p = N_A - N_D \approx N_A = 10^{17} cm^{-3} \tag{1}$$

We have the formula to determine the hole density if the we have the intrinsic carrier density (ni), the intrinsic potential (Ei) and the fermi level (Ef).

$$p = n_i \exp\left(\frac{E_i - E_F}{k_B T}\right) \tag{2}$$

Re-arrange equation (2), we can get

$$E_i - E_F = k_B T \ln\left(\frac{p}{n_i}\right) \tag{3}$$

The thermal energy $k_B T$ and the intrinsic carrier density n_i are given by the question as

$$k_B T = 0.026\,\mathrm{eV}, \tag{4}$$

$$n_i = 10^{10}\,\mathrm{cm}^{-3}. \tag{5}$$

Now, we take equation (1),(2),(3) into (4), we can get

$$E_i - E_F = k_B T \ln\left(\frac{p}{n_i}\right) = (0.026\,\mathrm{eV}) \ln\left(\frac{10^{17}}{10^{10}}\right) \approx 0.42\,\mathrm{eV}, \tag{6}$$

Finally we find that the fermi level is 0.42eV higher than Ei.

2 Question 2

A Schottky contact is formed by gold and n-type silicon. The doping level of silicon is $N_D = 10^{16}\,\mathrm{cm}^{-3}$. Intrinsic carrier density of silicon is $10^{10}\,\mathrm{cm}^{-3}$. What is the Fermi level in silicon?

We know gold has a work function ($q\phi_M$) = 5.1 eV. Silicon has an electron affinity (qχ) of 4.05 eV, and a permittivity of $\varepsilon = 11.7\,\varepsilon_0$, where $\varepsilon_0 = 8.854 \times 10^{-12}\,\mathrm{F\,m}^{-1}$. What is the depletion length in silicon?

Solution:

Similar to Q1, since n-type silicon and the dopant density N_D are given while the acceptor density N_A is not mentioned, we will take the dopant density as the electron density.

$$n = N_D - N_A \approx N_D = 1016 cm^{-3} \tag{1}$$

We have the formula to determine the electron density if the we have the intrinsic carrier density (ni), the intrinsic potential (Ei) and the fermi level (Ef).

$$n = n_i \exp\left(\frac{E_F - E_i}{k_B T}\right) \tag{2}$$

Re-arrange equation (2), we can get

$$E_F - E_i = k_B T \ln\left(\frac{n}{n_i}\right) \tag{3}$$

The thermal energy $k_B T$ and the intrinsic carrier density n_i are given by the question as

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$$n_i = 10^{10}\,\mathrm{cm}^{-3}. \tag{5}$$

Now, we take equation (1),(2),(3) into (4), we can get

$$E_F - E_i = k_B T \ln\left(\frac{n}{n_i}\right) = (0.026\,\text{eV}) \ln\left(\frac{10^{16}}{10^{10}}\right) \approx 0.36\,\text{eV}, \tag{6}$$

In a Schottky contact (Au on n-Si), we have the following physical property, where the built-in energy satisfies equation (7). The equation can be derive from corelation in figure 1.

$$q\phi_i = q\phi_M - q\chi - (E_C - E_F), \tag{7}$$

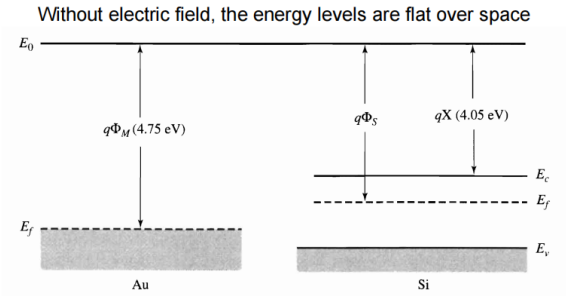


Figure 1: Energy level in a Schottky contact

Now, let's find the value of each parameter in equation (7). The physical properties of gold and silicon are provided in the question, as

$$q\,\phi_M = 5.1\,\text{eV}, \qquad \text{(gold work function)} \tag{8}$$

$$q\,\chi_{\text{Si}} = 4.05\,\text{eV}, \qquad \text{(silicon electron affinity)} \tag{9}$$

The energy difference between the conduction band and the Fermi level $E_C - E_F$ can be determined. Using $E_C - E_i = 0.56\,\text{eV}$ (the energy difference between the conduction band and the intrinsic Fermi level for Si at room temperature), we can obtain the built-in energy as

$$E_C - E_F = (E_C - E_i) - (E_F - E_i) = 0.56\,\text{eV} - 0.36\,\text{eV} = 0.20\,\text{eV}. \tag{10}$$

Taking (8),(9),(11) into (7), we will get

$$\phi_i = 5.1 - 4.05 - 0.20 = 0.85\,\text{V}. \tag{11}$$

There is another expression for the built-in potential based on the depletion width X_d, the permittivity ε, and the donor density N_D:

$$\phi_i = \frac{q\,N_D\,X_d^2}{2\,\varepsilon_{\text{Si}}}. \tag{12}$$

Rearranging (12), the depletion width is

$$X_d = \sqrt{\frac{2\,\varepsilon_{\text{Si}}\,\phi_i}{q\,N_D}}. \tag{13}$$

Last thing before the final result, we list all the material parameters below.

$$\varepsilon_{\text{Si}} = 11.7 \times \varepsilon_0 \tag{14}$$

$$\varepsilon_0 = 8.854 \times 10^{-12}\,\text{F/m} \tag{15}$$

$$N_D = 10^{16}\,\text{cm}^{-3} = 10^{22}\,\text{m}^{-3} \tag{16}$$

Note: uncommonly, we use meter as the unit of length here, due to the unit of ε_0
Finally, we determine the depletion length.

$$X_d = \sqrt{\frac{2\,\varepsilon_{\text{Si}}\,\phi_i}{q\,N_D}} \approx \sqrt{\frac{2(11.7)(8.854\times10^{-12})(0.85)}{(1.602\times10^{-19})(10^{22})}} \approx 3.3\times10^{-7}\,\text{m} = 0.33\,\mu\text{m}.$$

3 Question 3

A diode has a doping of $N_D = 10^{20}\,\mathrm{cm}^{-3}$ on the n-type side and $N_A = 10^{18}\,\mathrm{cm}^{-3}$ on the p-type side. What are the values of the depletion width (W_{d0}) and built-in voltage (ϕ_j)? What is the maximum electric field intensity? What are the depletion widths x_p, x_n?
What is the value of the depletion width (W_d) at a reverse bias of 5 V? At 25 V?

Solution:
Firstly, let's organize all the parameters that are known.

$$N_D = 10^{20}\,\mathrm{cm}^{-3} \quad \text{(donor density at n-side)} \tag{1}$$
$$N_A = 10^{18}\,\mathrm{cm}^{-3} \quad \text{(acceptor density at p-side)} \tag{2}$$
$$n_i = 10^{10}\,\mathrm{cm}^{-3} \quad \text{(intrinsic carrier density)} \tag{3}$$
$$V_T = 0.025\,\mathrm{V} \quad \text{(thermal voltage at room temp.)} \tag{4}$$
$$\varepsilon_{\mathrm{Si}} = 11.7 \times \varepsilon_0 = 11.7 \times 8.854 \times 10^{-12}\,\mathrm{F/m} \quad \text{(permittivity of Si)} \tag{5}$$
$$q = 1.6 \times 10^{-19}\,\mathrm{C} \quad \text{(charge of a single electron)} \tag{6}$$

We have a formula for the built-in voltage (ϕ_j), whose parameters are only N_D, N_A, n_i, and V_T. This formula is shown in equation (7).

$$\phi_j = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) \tag{7}$$

Taking (1),(2),(3),(4) into (7), we can determine the value of the built-in voltage of the pn junction.

$$\phi_j = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) = (0.025)\,\ln\left(\frac{10^{18}\cdot 10^{20}}{(10^{10})^2}\right) \approx 1.04\,\mathrm{V}. \tag{8}$$

Next, let's write down the formula for the pn junction depletion length (W_{d0}), where the parameters are $\varepsilon_{\mathrm{Si}}$, q, N_A, N_D and ϕ_j.

$$W_{d0} = \sqrt{\frac{2\,\varepsilon_{\mathrm{Si}}}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)\phi_j} \tag{9}$$

Take (1),(2),(6), (8) into (9), we can determined the depletion length. Please pay as much attention as you can to the UNITs!

$$\begin{aligned} W_{d0} &= \sqrt{\frac{2\,\varepsilon_{\mathrm{Si}}}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)\phi_j} \\ &= \sqrt{\frac{2(11.7)(8.854\times 10^{-14}\,\mathrm{F/cm})}{1.602\times 10^{-19}\,\mathrm{C}}\left(\frac{1}{10^{18}\,\mathrm{cm}^{-3}}+\frac{1}{10^{20}\,\mathrm{cm}^{-3}}\right)(1.04\,\mathrm{V})} \approx 0.0368\,\mu\mathrm{m}. \end{aligned} \tag{10}$$

To determine the maximum electric field intensity (E_{max}), we will apply the equation below which describes the correlation between the built-in voltage and E_{max}, where

$$V_{bi} = \tfrac{1}{2}\,E_{\max}\,W \tag{11}$$

Rearranging (11) and take (8),(10) into it, we can determine the maximum electric field intensity as

$$E_{\max} = \frac{2\phi_j}{W_{d0}} = \frac{2(1.04)}{0.0368}\ \tfrac{\mathrm{V}}{\mu\mathrm{m}} \approx 56.52\ \tfrac{\mathrm{V}}{\mu\mathrm{m}}. \tag{12}$$

At page 44 of the lecture slide, the formulas for the portion of the depletion width extended into p-side and n-side x_p, x_n are provided as

$$x_p = \frac{N_D}{N_A + N_D} W_{d0} \tag{13}$$

$$x_n = \frac{N_A}{N_A + N_D} W_{d0}. \tag{14}$$

Taking the values of N_A, N_D and W_{d0} into (13) and (14), x_p and x_n are determined as

$$x_p = \frac{10^{20}}{10^{18} + 10^{20}} \times 0.0368\,\mu\text{m} \approx 0.0364\,\mu\text{m}, \tag{15}$$

$$x_n = \frac{10^{18}}{10^{18} + 10^{20}} \times 0.0368\,\mu\text{m} \approx 0.00036\,\mu\text{m}. \tag{16}$$

Lastly, as shown in figure 2, for the pn junction with bias, we can determine the new built-in voltage using the formula in (17), where the bias V_R is the only variable.
Note: Conventionally, we define the positive end of V_R is connected to the n-side of the pn junction.

$$W_d = W_{d0}\sqrt{1 + \frac{V_R}{\phi_j}} \tag{17}$$

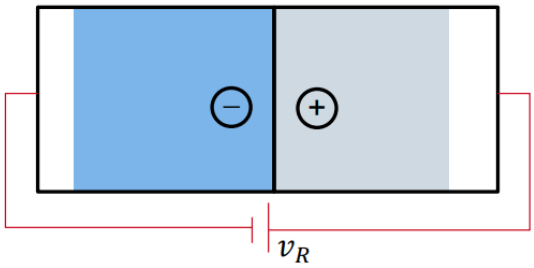


Figure 2: pn junction with bias

Using (17) with $V_R = 5\,\text{V}$,

$$W_d(5\,\text{V}) = W_{d0}\sqrt{1 + \frac{V_R}{\phi_j}} = (0.0368\,\mu\text{m})\sqrt{1 + \frac{5}{1.04}} \approx 0.0887\,\mu\text{m}.$$

Using (17) with $V_R = 25\,\text{V}$,

$$W_d(25\,\text{V}) = (0.0368\,\mu\text{m})\sqrt{1 + \frac{25}{1.04}} \approx 0.184\,\mu\text{m}.$$